①

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 11 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 2 & 0 & 5 & 0 & -4 & 27 \\ 3 & 2 & 7 & -6 & -8 & 38 \end{array}\right] \quad \begin{array}{l} -2R_1 + R_3 \longrightarrow R_3 \\ -3R_1 + R_4 \longrightarrow R_4 \\ \xrightarrow{\hspace{3cm}} \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 11 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 5 \\ 0 & 2 & 1 & -6 & -2 & 5 \end{array}\right] \quad \begin{array}{l} -2R_2 + R_4 \longrightarrow R_4 \\ \xrightarrow{\hspace{3cm}} \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & -2 & 11 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & 0 & 5 \end{array}\right] \quad \begin{array}{l} -R_3 + R_4 \longrightarrow R_4 \\ -2R_3 + R_1 \longrightarrow R_1 \\ \xrightarrow{\hspace{3cm}} \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 1 \\ 0 & 1 & 0 & -3 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$x_4 = a$

$x_5 = b$

$x_3 = 5$

$x_2 - 3a - b = 0$

$x_2 = 3a + b$

$x_1 - 2b = 1$

$x_1 = 2b + 1$

$(x_1, x_2, x_3, x_4, x_5) = \{(2b+1, 3a+b, 5, a, b)\}$ $a, b \in \mathbb{R}$

Non travial solution (more then solution)

For instance for any values of x_4 and x_5;

1) $(7, 6, 5, 1, 3)$ is a solution for $b=3$ and $a=1$

2) $(9, 13, 5, 3, 4)$ is a solution for $b=4$ and $a=3$

②

$$\begin{aligned} x_1+x_2+cx_3+x_4&=c \\ -x_2+x_3+2x_4&=0 \\ x_1+2x_2+x_3-x_4&=-c \end{aligned} \rightarrow \begin{bmatrix} 1 & 1 & c & 1 & c \\ 0 & -1 & 1 & 2 & 0 \\ 1 & 2 & 1 & -1 & -c \end{bmatrix} \xrightarrow[(-1)R_2 \to R_2]{R_3+(-1)R_1 \to R_3} \begin{bmatrix} 1 & 1 & c & 1 & c \\ 0 & 1 & -1 & -2 & 0 \\ 0 & 1 & (1-c) & -2 & -2c \end{bmatrix}$$

$$\xrightarrow{R_3+(-1)R_2 \to R_3} \begin{bmatrix} 1 & 1 & c & 1 & c \\ 0 & 1 & -1 & -2 & 0 \\ 0 & 0 & (2-c) & 0 & (-2c) \end{bmatrix} \xrightarrow{R_1+(-1)R_2 \to R_1} \begin{bmatrix} 1 & 0 & (c+1) & 3 & c \\ 0 & 1 & -1 & -2 & 0 \\ 0 & 0 & (2-c) & 0 & (-2c) \end{bmatrix}$$

a) If system has exactly one solution, then value of c;

$a_{33} \to$ 3. row and 3. column $\to (2-c) \neq 0$

$c \neq 2 \quad = \mathbb{R} - \{2\}$

b) If system has infinite number of solution, then value of c;

$\left.\begin{aligned} \text{I)}\ & 2-c=0 \\ \text{II)}\ & -2c=0 \end{aligned}\right\}$ should be, but there is no $c \in \mathbb{R}$ to satisfy the equation, so solution set is empty $= \emptyset$

Option b is not possible.

c) If system has no solution then values of c;

$\left.\begin{aligned} \text{I)}\ & 2-c=0 \\ \text{II)}\ & -2c \neq 0 \end{aligned}\right\}$ $\underline{c=2}$

③

$$x+2y-3z=4$$
$$3x-y+5z=2$$
$$4x+y+(a^2-14)z=a+2$$

$$\to \left[\begin{array}{cccc} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & (a^2-14) & (a+2) \end{array}\right] \xrightarrow[R_3-4R_1\to R_3]{R_2-3R_1\to R_2} \left[\begin{array}{cccc} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & 10 \\ 0 & -7 & (a^2-2) & (a-14) \end{array}\right]$$

$$\frac{R_2}{-7}\to R_2 \quad \left[\begin{array}{cccc} 1 & 2 & -3 & 4 \\ 0 & 1 & -2 & 10/7 \\ 0 & -7 & (a^2-2) & a-14 \end{array}\right] \xrightarrow{R_3+7R_2\to R_3} \left[\begin{array}{cccc} 1 & 2 & -3 & 4 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & a^2-16 & a-4 \end{array}\right]$$

* if system has no solition then a;

$$\left.\begin{array}{ll} \text{I) } a^2-16=0 & \to a=\{-4,4\} \\ \text{II) } a-4\neq 0 & \to a\neq 4 \end{array}\right\} a=-4$$

* if system has exactly one solition then a;

$a^2-16\neq 0$

$a\neq 4,-4 \quad \to R-\{-4,4\}$

* if system has infinitely many solition then a;

$$\left.\begin{array}{ll} \text{I) } a^2-16=0 & \Rightarrow a=-4,4 \\ \text{II) } a-4=0 & \Rightarrow a=4 \end{array}\right\} a=4$$

(4)

$$\begin{bmatrix}2&1&0\\0&1&-1\\-2&1&1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\2\\-2\end{bmatrix}$$

$$\begin{bmatrix}2&1&0\\0&1&-1\\-2&1&1\end{bmatrix}\xrightarrow[E_1]{R_1+R_3\to R_3}\begin{bmatrix}2&1&0\\0&1&-1\\0&2&1\end{bmatrix}\xrightarrow[E_2]{-2R_2+R_3\to R_3}$$

$$\begin{bmatrix}2&1&0\\0&1&-1\\0&0&3\end{bmatrix}=A\,E_1\,E_3=U$$

$$E_1=\begin{bmatrix}1&0&0\\0&1&0\\1&0&1\end{bmatrix}\qquad E_2=\begin{bmatrix}1&0&0\\0&1&0\\0&-2&1\end{bmatrix}$$

$$E_1^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\-1&0&1\end{bmatrix}\qquad E_2^{-1}=\begin{bmatrix}1&0&0\\0&1&0\\0&2&1\end{bmatrix}\to E_1^{-1}.E_2^{-1}=L$$

$$L=\begin{bmatrix}1&0&0\\0&1&0\\-1&2&1\end{bmatrix}\qquad A=L.U\begin{bmatrix}o_1\\o_2\\o_3\end{bmatrix}$$

$$\begin{bmatrix}1&0&0\\0&1&0\\-1&2&1\end{bmatrix}\cdot\begin{bmatrix}2&1&0\\0&1&-1\\0&0&3\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\2\\-2\end{bmatrix}$$

$$\begin{bmatrix}1&0&0\\0&1&0\\-1&2&1\end{bmatrix}\begin{bmatrix}o_1\\o_2\\o_3\end{bmatrix}=\begin{bmatrix}1\\2\\-2\end{bmatrix}$$

$o_1=1$

$o_2=2$

$o_3=-5$

$$\begin{bmatrix}2&1&0\\0&1&-1\\0&0&3\end{bmatrix}\cdot\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\2\\-5\end{bmatrix}$$

$\longrightarrow$ $3z=-5$ $\quad$ $y+\frac{5}{3}=2$ $\quad$ $2x+\frac{1}{3}=1$

$z=-\frac{5}{3}$ $\quad$ $y=1/3$ $\quad$ $x=\frac{1}{3}$

$$\boxed{\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1/3\\1/3\\-5/3\end{bmatrix}}$$

(5)

$$A \rightarrow \left[\begin{array}{cccc|cccc} 2 & 0 & -1 & 3 & 1 & 0 & 0 & 0 \\ 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 4 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 3 & -2 & -5 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_1}$$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 2 & 0 & -1 & 3 & 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 3 & -2 & -5 & 0 & 0 & 0 & 1 \end{array}\right] \quad \begin{array}{l} -2R_1 + R_2 \rightarrow R_2 \\ -4R_1 + R_3 \rightarrow R_3 \\ \longrightarrow \\ -R_1 + R_4 \rightarrow R_4 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 0 & 9 & -12 & -5 & 0 & -4 & 1 & 0 \\ 0 & 5 & -5 & -6 & 0 & -1 & 0 & 1 \end{array}\right] \quad \begin{array}{l} -9/4\, R_2 + R_3 \rightarrow R_3 \\ \longrightarrow \\ -5/4\, R_2 + R_4 \rightarrow R_4 \end{array}$$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 0 & 0 & 15/4 & -29/4 & -9/4 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 15/4 & -29/4 & -5/4 & 6/4 & 0 & 1 \end{array}\right] \quad -R_3 + R_4 \rightarrow R_4$$

$$\left[\begin{array}{cccc|cccc} 1 & -2 & 3 & 1 & 0 & 1 & 0 & 0 \\ 0 & 4 & -7 & 1 & 1 & -2 & 0 & 0 \\ 0 & 0 & 15/4 & -29/4 & -9/4 & 1/2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right]$$

Matrix A has no inverse

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$$B \rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & -3 & 0 & 1 & 0 & 0 \\ 0 & 2 & -3 & 7 & 0 & 0 & 1 & 0 \\ 2 & 1 & -2 & 12 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\substack{R_1 + R_2 \to R_2 \\ -2R_1 + R_4 \to R_4}}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & -3 & 7 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 & -2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{\substack{-2R_2 + R_3 \to R_3 \\ -R_2 + R_4 \to R_4}}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 3 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & -3 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 + R_4 \to R_4}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 3 & 2 & 2 & -1 & 0 \\ 0 & 0 & 0 & 3 & -5 & -3 & 1 & 1 \end{array}\right] \xrightarrow[\frac{1}{3} R_4 \to R_4]{-R_3 \to R_3}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 5 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -3 & 2 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & -5/3 & -1 & 1/3 & 1/3 \end{array}\right] \xrightarrow[-5R_4 + R_1 \to R_1]{\substack{3R_4 + R_3 \to R_3 \\ -2R_4 + R_2 \to R_2}}$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & -1 & 0 & 28/3 & 5 & -5/3 & -5/3 \\ 0 & 1 & -1 & 0 & 13/3 & 3 & -2/3 & -2/3 \\ 0 & 0 & 1 & 0 & -3 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -5/3 & -1 & 1/3 & 1/3 \end{array}\right] \xrightarrow[R_3 + R_1 \to R_1]{R_3 + R_2 \to R_2}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & \\ 0 & 1 & 0 & 0 & \\ 0 & 0 & 1 & 0 & \\ 0 & 0 & 0 & 1 & \end{array}\right. \underbrace{\begin{array}{cccc} 19/3 & 4 & -5/3 & -2/3 \\ 4/3 & 2 & -2/3 & 1/3 \\ -3 & -1 & 0 & 1 \\ -5/3 & -1 & 1/3 & 1/3 \end{array}}_{B^{-1}} \left.\right]$$

⑥

Find inveres of matribes buy using elimantary matribes.

$$A=\begin{bmatrix}1 & 0 & -1\\ 0 & 6 & -1\\ 0 & 0 & 4\end{bmatrix} \qquad B=\begin{bmatrix}1 & 0 & -2\\ 0 & 2 & 1\\ 0 & 0 & 1\end{bmatrix}$$

$$A=\begin{bmatrix}1 & 0 & -1\\ 0 & 6 & -1\\ 0 & 0 & 4\end{bmatrix} \quad \underbrace{\frac{R_3}{4}+R_2 \to R_2}_{E_1},\ \underbrace{\frac{R_3}{4}+R_1 \to R_1}_{E_2} \quad \begin{bmatrix}1 & 0 & 0\\ 0 & 6 & 0\\ 0 & 0 & 4\end{bmatrix} \quad \underbrace{\frac{R_2}{6} \to R_2}_{E_3},\ \underbrace{\frac{R_3}{4} \to R_3}_{E_4} \quad \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$E_1=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \frac{R_3}{4}+R_2 \to R_2 \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1/4\\ 0 & 0 & 1\end{bmatrix} E_1$$

$$E_2=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \frac{R_3}{4}+R_1 \to R_1 \begin{bmatrix}1 & 0 & 1/4\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix} E_2$$

$$E_3=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \frac{R_2}{6} \to R_2 \begin{bmatrix}1 & 1 & 0\\ 0 & 1/6 & 0\\ 0 & 0 & 1\end{bmatrix} E_3$$

$$E_4=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} \frac{R_3}{4} \to R_3 \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1/4\end{bmatrix} E_4$$

$$A^{-1} = E_4 . E_3 . E_2 . E_1 . I \,//$$

⑥ devamı

$$B = \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \quad \tfrac{1}{2} R_2 \to R_2$$

$$\downarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[2R_3 + R_1 \to R_1]{-\frac{1}{2} R_3 + R_2 \to R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 0 & 1/2 & -1/2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

where the right block is B^{-1}.

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B^{-1} = E_3 . E_2 . E_1$$

$$B^{-1} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1/2 & -1/2 \\ 0 & 0 & 1 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & a & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} ab+1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} ab+1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{bmatrix} = \begin{bmatrix} ab+1 & a & 0 \\ b & 1 & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} ab+1 & a & 0 & 1 & 0 & 0 \\ b & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & c & 0 & 0 & 1 \end{array}\right] \xrightarrow{-aR_2 + R_1 \to R_1}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -a & 0 \\ b & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & c & 0 & 0 & 1 \end{array}\right] \xrightarrow{1/c\, R_3 \to R_3}$$

$$\underbrace{\begin{matrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}}_{I} \; \Bigg| \; \underbrace{\begin{matrix} 1 & -a & 0 \\ -b & ab+1 & 0 \\ 0 & 0 & 1/c \end{matrix}}_{A^{-1}}$$

⑧ Let $p(x) = a_0 + a_1 x + a_2 x^2$ be the polynomial which passes through $(1,4), (2,0), (3,12)$

So all these points should satisfy $p(x)$

Now taking $(1,4)$;

We have $x = 1$, $p(x) = 4$

$$4 = a_0 + a_1(1) + a_2(1)^2$$

$$\rightarrow \quad = a_0 + a_1 + a_2$$

Taking $(2,0)$ $\quad x = 2$, $p(x) = 0$

$$\rightarrow 0 = a_0 + 2a_1 + 4a_2$$

Taking $(3,12)$ $\quad x = 3$, $p(x) = 12$

$$\rightarrow \quad 12 = a_0 + 3a_1 + 9a_2$$

We have three equation

$$a_0 + a_1 + a_2 = 4$$

$$a_0 + 2a_1 + 4a_2 = 0$$

$$a_0 + 3a_1 + 9a_2 = 12$$

Above system linear equation with augumented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 0 \\ 1 & 3 & 9 & 12 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & -4 \\ 1 & 3 & 9 & 12 \end{array}\right] \quad R_2 \rightarrow R_2 - R_1$$

8

devamı

$$B = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow[\underbrace{2R_3 + R_1 \to R_1}_{E_2}]{\overbrace{-R_3 + R_2 \to R_2}^{E_1}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\overbrace{\frac{R_2}{2} \to R_2}^{E_3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{-R_3 + R_2 \to R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \to E_1$$

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{2R_3 + R_1 \to R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \to E_2$$

$$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\frac{R_2}{2} \to R_2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \to E_3$$

$$\boxed{B^{-1} = E_3 . E_2 . E_1 . I}$$

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$$200 + x_1 - x_4 - x_2 = 0$$
$$x_4 + x_5 - x_6 = 0$$
$$100 + x_6 - x_1 - x_3 = 0$$
$$-300 - x_5 + x_3 + x_2 = 0$$

$\rightarrow$

$$200 = x_2 + x_4 - x_1$$
$$0 = x_4 + x_5 - x_6$$
$$100 = x_1 + x_3 - x_6$$
$$300 = x_2 + x_3 - x_5$$

$$\begin{array}{cccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & \\ \end{array}$$
$$\left[\begin{array}{cccccc|c} -1 & 1 & 0 & 1 & 0 & 0 & 200 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 & 0 & -1 & 100 \\ 0 & 1 & 1 & 0 & -1 & 0 & 300 \end{array}\right]$$

$$\left[\begin{array}{cccccc|c} -1 & 1 & 0 & 1 & 0 & 0 & 200 \\ 1 & 0 & 1 & 0 & 0 & -1 & 100 \\ 0 & 1 & 1 & 0 & -1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & -3 & 0 \end{array}\right]$$

$R_1 + R_2 \rightarrow R_2$

$$\left[\begin{array}{cccccc|c} -1 & 1 & 0 & 1 & 0 & 0 & 200 \\ 0 & 1 & 1 & 1 & 0 & -1 & 300 \\ 0 & 1 & 1 & 0 & -1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right]$$

$-R_2 + R_3 \rightarrow R_3$

$-R_3 \rightarrow R_3$

$$\left[\begin{array}{cccccc|c} -1 & 1 & 0 & 1 & 0 & 0 & 200 \\ 0 & 1 & 1 & 1 & 0 & -1 & 300 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right]$$

$-x_1 + x_2 + x_4 \rightarrow 200$

$x_2 + x_3 + x_4 - x_6 \rightarrow 300$

$x_4 + x_5 - x_6 \rightarrow 0$

$$\boxed{\begin{array}{l} x_1 = 200 - x_2 - x_6 + x_5 \\ x_3 = x_5 - x_2 + 300 \\ x_4 = x_6 - x_5 \end{array}}$$

$x_3 = \boxed{1000}$

$x_5 = x_6 = \boxed{50}$

$x_4 \boxed{= 0}$

$x_1 = 200 + 650 - 50 + 50 \boxed{= 850}$

$x_2 \boxed{= -650}$

10

$400 + x_2 = x_1$

$600 + x_4 = x_1 + x_3$

$300 = x_2 + x_3 + x_5$

$100 = x_5 + x_4$

$$\begin{array}{c|c}
 & \begin{array}{ccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & \end{array} \\
\begin{array}{c} R_1 \\ R_2 \\ R_3 \\ R_4 \end{array} & \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & 400 \\ 1 & 0 & 1 & -1 & 0 & 600 \\ 0 & 1 & 1 & 0 & 1 & 300 \\ 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right]
\end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & 400 \\ 0 & 1 & 1 & -1 & 0 & 200 \\ 0 & 1 & 1 & 0 & 1 & 300 \\ 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right] \quad \begin{array}{l} R_2 = R_1 - R_2 \\ R_2 = -R_2 \end{array}$$

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & 0 & 0 & 400 \\ 0 & 1 & 1 & -1 & 0 & 200 \\ 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right] \quad \begin{array}{l} R_3 = R_2 - R_3 \\ R_3 = -R_3 \end{array}$$

$x_1 - x_2 = 400$

$x_2 + x_3 - x_4 = 200$

$x_4 + x_5 = 100$

$x_5 = k$

$x_4 = 100 - k$

$x_2 = L$

$x_3 = 200 + 100 - k - L$

$= 300 - k - L$

$x_1 = 400 + L$

From Q. $\boxed{x_5 = 1000\ (k)}$ $\boxed{x_3 = 0}$

$\boxed{x_2 = -700}$ $\boxed{x_1 = -300}$ $\boxed{x_4 = -900}$

(11)

Solve equations (1), (2) and (3)

$I_a = -2A$

$I_b = 2A$

$I_c = -3A$

The broch currents are,

Current I_1 is;

$I_1 = -I_a$

$= -(-2)$

$= 2A$

Current I_2 is;

$I_2 = I_b - I_a$

$= 2 - (-2)$

$= 4A$

Current I_3 is;

$I_3 = I_b$

$= 2A$

Current I_4 is;

$I_4 = I_b$

$= 2A$

Current I_5 is;

$I_5 = I_b - I_c$

$= 2 - (-3)$

$= 5A$

Current I is;

$I_6 = -I_c$

$= -(-3)$

$= 3A$